Resonance Tree — Solar Lighting Study

Phase 1 (trunk + root lights) & Phase 2 (hanging canopy lights) · Black Rock City, Aug 30 – Sep 8 2026 · prepared 2026-07-17 (rev. v5 — calibrated against Ben's bench measurements) · geometry-accurate ray-traced study (~500,000 sun rays vs the woven structure)

Bottom line. Every light is a **standalone island**, and the whole chain is now calibrated against Ben's bench measurements (Voltaic panel INA tests + the July field-cycle ledger). A "5 W" panel really delivers **3.85 W panel-side** in full sun, ≈3.2 W into the battery at peak, and ×0.63 of panel energy survives the daily charge chain. Result: the median canopy light banks **≈10.5 Wh/day into its own cell** — about **7.7 h of full-brightness RGBW per night** at the measured 1.36 W battery-side draw (dark window is ~10 h; dimmed it covers the night). Nine lights sit under 4 h, and the six inner-south lights get minutes — relocate them. Rated 5 W is never the operating point.

Method — measured, not estimated

The real tree geometry (Rhino-derived woven model) was geolocated in SketchUp and verified against the almanac (solar noon 12:57 PDT, altitude 56.9°, noon shadow due north). Every 10 minutes across all 10 event days, rays were cast from a grid of points on each panel's face toward the sun; hits against any strut count as shade, rays through weave gaps count as sun — so dappled light is measured, not guessed. Panel positions: 16 ground/trunk panels as placed in the design model; 78 canopy lights at the exact hanging positions from the lighting system's fixtures.json.

Model v5 — calibrated on Ben's measurements (repo: beneckart/resonance-lighting). Panel truth from the 2026-06-29 panel-side INA test (ADR 0026): Voltaic P105 "5 W" → **3.85 W real full-sun panel-side** (×0.77 nameplate, fixed charge setpoint), charger input ×0.90, battery-side ≈3.2 W peak. Daily charge chain from the July field-cycle ledger: best sunny day 9.02 Wh input → 5.7 Wh into the cell = **×0.63** (conversion + taper + system overhead). On top of that: my ray-traced geometric shading (lit-fraction² ×0.75 when partially shaded), playa-excess heat (calculated), dust ×0.95. LED draw also measured: full RGBW = 1.364 W battery-side, not the 4 W optical rating. Panel roles per ADR 0026: P105 5 W on the downlights and uplight wings — exactly this study's 94 lights. Cross-check: model median canopy = 2.1 Wh per nameplate-watt per day battery-side vs Ben's flat-panel California bench at 2.85 — the model sits appropriately below his clean-bench number (playa heat, dust, weave dapple).

Phase 1 — trunk ring & root lights (16)

12 lights on the lower ring with panels at the outer bark (following the trunk's flare), 4 on the root arms. All panels set to their per-position optimum angle (the hinge targets).

Group	Count	Real Wh/day each	Notes
Roots (RT-SSW 27.5, RT-SE 25.8)	4	18 – 27.5	Most reliable; 35° south tilt
Ring, S/SW/SE side	≈5	18 – 26	Rides the southern sun all day
Ring, W/E sides	≈5	12 – 18	Half-day sun (morning or evening)
Ring, N side	≈3	9 – 13	Weakest; aimed ESE/WSW to catch shoulder sun

Ground subtotal: **160 Wh/day into batteries** (254 panel-DC). Recommended angles are in the study spreadsheet ("Aim tilt / Aim facing" columns) — rule of thumb: south side 35° facing S; east side 40–60° facing ESE–SE; west/north side 45–65° facing WSW.

Phase 2 — hanging canopy lights (78)

Lights hang at their designed positions, 0.06–2.84 m below the branches (average measured drop 0.94 m), panels flat on each lantern top. Named by ring + bearing (e.g. **CL-O12 157°SSE** = outer ring, light 12, bearing 157°). Rings: inner ≤ 2.7 m radius (28), middle 2.7–5 m (26), outer > 5 m (24).

Ring	Count	Battery Wh/day (min / median / max)	Runtime @1.36 W full RGBW (median)
Outer (canopy edge)	24	battery: 9.2 / 11.1 / 13.4	≈8.1 h
Middle	26	battery: 6.9 / 10.7 / 12.9	≈7.9 h
Inner (under trunk canopy)	28	battery: **0.3** / 6.7 / 12.2	≈4.9 h

The canopy underside is generous with sun: even hanging under the branches, edge and mid-ring lights see 60–95 % open sky and clear 18+ Wh/day. Performance decays toward the trunk, and collapses on the inner **south** arc where a light's own tree blocks the entire southern sun path.

Hang-height findings & the 7-foot rule

Constraint applied: **panel bottom no lower than 7 ft (2.13 m)** for head clearance. Each light was re-tested at candidate hang heights between its current position and the 7 ft floor:

- **54 of 78 lights are already at a good height** — lowering them buys <2 % sky view; leave the cords as designed.
- **24 lights gain by hanging lower** (mostly inner/middle ring on the N–E arc, currently 3.4–3.9 m high, tucked close under dense weave). Recommended heights are per-light in the spreadsheet ("rec height" — typically 2.1–2.6 m). Longer cords = more side-sky under the canopy skirt. Estimated gain 5–20 % each.
- **Six lights sit below 7 ft entirely** — CL-I11/12/14/16/17/18 (inner ring, 165°–201°, S–SSW), hanging at 0.05–0.94 m. These are the same six worst performers (0.4–2.5 Wh/day): low + deep + south-side = triple penalty. **Decision needed (no wiring exists between lights, so remote panels are NOT an option):** (a) accept them as near-dark accents (6–30 min of light), or (b) relocate those six lights outward/upward to open-sky spots — moving a light IS moving its power plant.

Recommendations

- **Proceed with the canopy-edge design** — outer + middle rings are solid solar citizens (≈1,060 Wh/day between them).
- **Lengthen 24 cords** per the spreadsheet's rec-height column (respecting the 7 ft floor).
- **Relocate the six inner-south lights** — with standalone lights the only fix is moving them; they are unsalvageable in place (0.1–0.5 h of runtime).
- **Ground panels:** keep the optimized aims; the four root panels and the S/SW ring arc carry that subsystem.
- **Budget per light, not per system:** median canopy light banks ≈10.5 Wh/day → ≈7.7 h of full-brightness RGBW (measured 1.36 W draw) vs the ~10 h dark window: full-power all night doesn't quite close; a modest dim profile does. This matches Ben's own field-cycle conclusion (thin positive margin) — two independent methods now agree.

- Verify the assumed 10×7″ 5 W panel spec for the lanterns; all numbers rescale linearly with panel wattage.

Deliverables: interactive 3D viewer (canopy layer toggleable) · study spreadsheet (per-light data, per-10-min output, shading-vs-orientation loss split) · SketchUp model with all 94 lights movable · datasets in git + SUNEAST. Sun/shade data: blender-architect study, 2026-07-16/17.